

Mendelian Susceptibility to Mycobacterial Disease due to Complete IL-12R β 1 Deficiency — Comprehensive Disease Report

Disease: Mendelian Susceptibility To Mycobacterial Diseases Due To Complete IL12RB1 Deficiency **MONDO ID:** MONDO:0013955 · **Gene:** *IL12RB1* (HGNC:5971) · **Locus:** 19q13.11 · **Inheritance:** Autosomal recessive **Category:** Mendelian inborn error of immunity (inborn error of IFN- γ immunity)

Summary

Complete IL-12 receptor β 1 (IL-12R β 1) deficiency is an autosomal-recessive inborn error of immunity and the **single most common genetic cause of Mendelian Susceptibility to Mycobacterial Disease (MSMD)**. It is caused by biallelic loss-of-function variants in *IL12RB1* (19q13.11), which encodes the β 1 chain shared by the receptors for interleukin-12 (IL-12) and interleukin-23 (IL-23). Loss of this chain simultaneously abolishes IL-12-driven IFN- γ production by T and NK cells and IL-23-driven IL-17 immunity. The functional consequence is a narrow but characteristic clinical syndrome: childhood-onset disease caused by weakly virulent mycobacteria — chiefly *Mycobacterium bovis* BCG (vaccine strain) and environmental/non-tuberculous mycobacteria — and by non-typhoidal *Salmonella*, with mucocutaneous candidiasis in roughly one quarter of patients ([PMID: 21057261](#), [PMID: 24186907](#)).

The largest characterization to date — an international survey of 141 patients from 102 kindreds across 30 countries — established that IL-12R β 1 deficiency has a **comparatively favorable prognosis** (70% survival, mean age at follow-up 12.7 $\pm$ 9.8 years) and **incomplete clinical penetrance** (27% of genetically affected siblings remained asymptomatic) ([PMID: 21057261](#)). Mechanistically the disorder is classified among MSMD genes that impair the

production of IFN- γ (with *IL12B*, *IRF8*, *ISG15*, *NEMO*), as opposed to those that impair the **response** to IFN- γ (*IFNGR1*, *IFNGR2*, *STAT1*, etc.) — a distinction that explains why adjunctive recombinant IFN- γ , which bypasses the receptor block, is clinically useful (PMID: 25453225).

A crucial diagnostic caveat is that affected individuals are **otherwise healthy with normal routine hematological and immunological tests**; diagnosis therefore requires targeted functional assays (defective cellular responses to IL-12; absent IL-12R β 1 surface expression) and genetic confirmation (PMID: 25453225, PMID: 11424023). Management centers on prolonged pathogen-directed antimycobacterial/antibacterial therapy, adjunctive recombinant IFN- γ , avoidance of the live BCG vaccine, and hematopoietic stem cell transplantation (HSCT) for severe or refractory disease, which achieves markedly higher antimycobacterial cure rates (PMID: 31367980, PMID: 41748971). An *Il12rb1*-deficient mouse recapitulates disseminated BCG susceptibility and is being used to dissect the residual, IFN- γ -independent protective immunity that underlies the disorder's low penetrance (PMID: 35891311).

1. Disease Information

Overview. Complete IL-12R β 1 deficiency is an isolated (non-syndromic) inborn error of IFN- γ immunity that produces selective predisposition to clinical disease from weakly virulent intracellular pathogens — principally mycobacteria (BCG, non-tuberculous mycobacteria [NTM], and, less often, *M. tuberculosis*) and non-typhoidal *Salmonella* — in individuals who are otherwise healthy and immunocompetent against most other microbes (PMID: 12591909, PMID: 21057261). It is the most common of the ~21–31 recognized MSMD genetic etiologies.

Key identifiers. | Resource | Identifier | |---|---| | MONDO | MONDO:0013955 | | OMIM (phenotype) | #614891 (Immunodeficiency 30 / MSMD, IL12RB1) | | OMIM (gene) | *IL12RB1* 601604 | | Gene / HGNC | *IL12RB1* / HGNC:5971 | | Orphanet | MSMD (disease group) | | MeSH | Related to "Mycobacterium Infections" / primary immunodeficiency terms | | ICD-10 | D84.8/D84.9 (other/unspecified immunodeficiency) | | ICD-11 | 4A00.x (inborn errors of immunity) |

Synonyms / alternative names. IL-12R β 1 deficiency; interleukin-12 receptor β 1 chain deficiency; IL12RB1 deficiency; MSMD due to IL-12R β 1 deficiency; Immunodeficiency 30.

Information source. The disease-level knowledge here derives predominantly from **aggregated disease-level resources and cohort studies** — most notably the international 141-patient survey (PMID: 21057261), the earlier 41-patient cohort (PMID: 12591909), and multiple national cohorts (Iran, Mexico, China) — supplemented by individual case reports.

2. Etiology

Disease causal factor — genetic. The disease is monogenic: biallelic (homozygous or compound-heterozygous) loss-of-function variants in *IL12RB1* that abolish surface expression of the receptor $\beta 1$ chain and eliminate cellular responsiveness to IL-12 (and IL-23). Direct proof of causality comes from complementation: "*Transfection of the patients' T cells with wild-type IL12RB1 restored IL-12Rbeta1 expression and function*" (PMID: 11424023).

Genetic risk factors. The causal variants are the risk factor. Because the disorder is recessive, the principal population-level risk amplifiers are **consanguinity** and **founder alleles**. Parental consanguinity is repeatedly documented in MSMD/IL-12R $\beta 1$ cohorts from high-consanguinity regions (e.g., Iran, Turkey) (PMID: 32602053, PMID: 30968642). The closely related IL-12p40 (*IL12B*) deficiency demonstrates recurrent founder alleles affecting 25 of 30 kindreds, supporting an analogous pattern for *IL12RB1* (PMID: 23429356).

Environmental risk factor — the essential trigger. Genotype alone is insufficient to cause overt disease; an **environmental/infectious exposure is the obligate second hit**. The dominant trigger is **BCG vaccination** (live attenuated *M. bovis*): among vaccinees in the near-phenocopy IL-12p40 deficiency, BCG disease occurred in 97.5% (PMID: 23429356); in the international IL-12R $\beta 1$ cohort, BCG was the most frequent causative organism (n=65) (PMID: 21057261). Environmental non-tuberculous mycobacteria and *Salmonella* exposure are the other triggers.

Protective factors. No specific protective allele is established. However, **residual IFN- γ -independent antimycobacterial immunity** clearly protects a subset of genetically affected individuals, accounting for asymptomatic siblings and the low recurrence rate (PMID: 35891311, PMID: 21057261). At the public-health level, **avoiding BCG vaccination** in at-risk families is strongly protective against the most common disease trigger.

Gene–environment interaction. This disorder is a paradigm of gene × environment interaction: a fixed germline lesion (loss of IL-12/IL-23 signaling) becomes clinically manifest only upon encounter with specific weakly virulent intracellular pathogens, while the same individuals resist most other microbes ([PMID: 12591909](#)).

3. Phenotypes

The clinical picture is dominated by infection. Frequencies below are drawn chiefly from the 141-patient international survey ([PMID: 21057261](#)) and the candidiasis series ([PMID: 24186907](#)).

Phenotype	Type	Onset	Frequency	Suggested HPO term
BCG disease (BCGitis/BCGosis, local→disseminated)	Clinical sign / infection	Infancy (post-vaccination)	Most common organism (n=65 of causative isolates)	HP:0002841 (Recurrent mycobacterial infections); HP:0032262 (BCG vaccine adverse event)
Non-tuberculous / environmental mycobacterial disease	Infection	Childhood	n=9	HP:0002841
<i>M. tuberculosis</i> disease	Infection	Childhood	n=4	HP:0032256 (Tuberculosis)
Non-typhoidal <i>Salmonella</i> infection (often disseminated)	Infection	Childhood	22 of the 24 non-mycobacterial probands	HP:0002718 (Recurrent bacterial infections)
Mucocutaneous candidiasis (mostly oropharyngeal)	Infection / mucosal sign	Median 1.5 yr (earliest infection type)	~23–25% (33/141)	HP:0002728 (Chronic mucocutaneous candidiasis)
Lymphadenitis / lymphadenopathy	Clinical sign	Childhood	Common	HP:0002716 (Lymphadenopathy)
Hepatomegaly / hepatosplenic involvement	Clinical sign	Childhood	Common in disseminated disease	HP:0002240 (Hepatomegaly)
Fever, failure to thrive / growth retardation	Symptom / sign	Childhood	Frequent with disseminated disease	HP:0001945; HP:0001510

Onset. First infection occurs at a **mean age of 2.4 years**; candidiasis, when present, tends to be the earliest manifestation (median 1.5 yr) ([PMID: 21057261](#), [PMID: 24186907](#)).

Severity / progression. Highly **variable** — ranging from localized BCGitis to fatal disseminated disease. A hallmark is that **mycobacterial infections generally do not recur** after successful treatment, distinguishing this from IFN- γ -receptor defects (PMID: 12591909); *Salmonella*, by contrast, recurs more often (in the parallel IL-12p40 disorder, salmonellosis recurred in 36.4% vs 25% mycobacterial recurrence) (PMID: 23429356).

Quality-of-life impact. Not formally measured with standardized instruments (EQ-5D/SF-36) in the literature reviewed. Morbidity derives from repeated hospitalizations, prolonged multidrug therapy, surgical drainage of lymphadenitis, and, in severe cases, disseminated organ involvement.

4. Genetic / Molecular Information

Causal gene. *IL12RB1* (OMIM 601604; HGNC:5971), 19q13.11, encoding the IL-12 receptor $\beta 1$ chain shared by the IL-12 (IL-12R $\beta 1$ /IL-12R $\beta 2$) and IL-23 (IL-12R $\beta 1$ /IL-23R) receptor complexes.

Variant classification and types. Reported pathogenic variants are diverse and include **missense, nonsense, frameshift, and splice-site** alleles. Examples from the reviewed literature: - Homozygous **missense** preventing receptor expression and abolishing IL-12 responses (PMID: 11424023). - **Nonsense** Q285X (c.853C>T, exon 9) causing loss of surface expression (PMID: 19099833). - **Splice-site** c.783+1G>A (homozygous), disease-causing (PMID: 30968642). - Homozygous missense or nonsense alleles in three Iranian patients that "*caused the IL-12R $\beta 1$ protein not to be expressed on the cell membrane and completely abolished the cellular response to recombinant IL-12*" (PMID: 29256176).

All are classified **pathogenic** under ACMG/AMP criteria on the basis of null functional effect, segregation, and rarity.

Functional consequence. Complete loss of function (null) — absent surface receptor and abolished IL-12/IL-23 signaling. There is no gain-of-function or dominant-negative mechanism; heterozygous carriers are asymptomatic.

Origin. Germline, biallelic. No somatic contribution.

Allele frequency. Individual pathogenic alleles are very rare in gnomAD; disease arises through recessive inheritance, enriched by consanguinity and country-specific founder alleles (PMID: 23429356, PMID: 32602053).

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier gene or epigenetic mechanism is established. The disorder is a point-lesion Mendelian condition, not a chromosomal/structural disorder. The residual protective immunity driving low penetrance is likely determined by redundant, partly IFN- γ -independent pathways rather than a single modifier locus (PMID: 35891311).

5. Environmental Information

Infectious agents (central to the disease). - *Mycobacterium bovis* BCG (live vaccine strain) — the dominant trigger and most frequent isolate (PMID: 21057261). - **Non-tuberculous / environmental mycobacteria** (e.g., *M. avium* complex) — PMID: 23726680. - *Mycobacterium tuberculosis* — less common but reported. - **Non-typhoidal *Salmonella*** (e.g., *S. Typhimurium*), frequently disseminated (PMID: 41771439). - *Candida* species — mucocutaneous candidiasis in ~25% (PMID: 24186907). - More rarely other intramacrophagic bacteria, fungi (e.g., *Histoplasma*), parasites, and perhaps a few viruses (PMID: 25453225, PMID: 36044171).

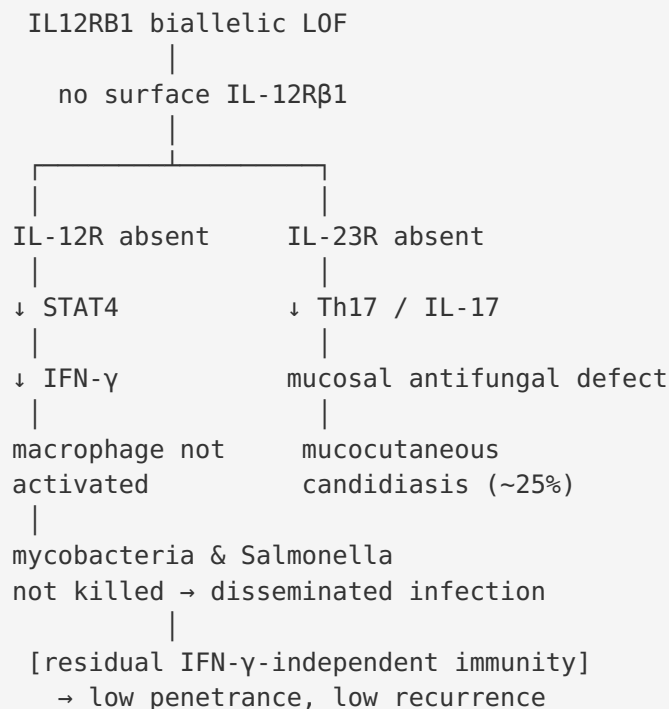
Non-infectious environmental / lifestyle factors. None causative. The single most consequential "environmental" exposure is a medical intervention — **BCG vaccination** — which precipitates disease in the majority of vaccinated patients. No toxin, occupational, or dietary factor is implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in *IL12RB1*** (missense/nonsense/frameshift/splice) **leads to** absent or non-functional IL-12R β 1 chain on the cell surface (PMID: 11424023, PMID: 29256176).
2. Loss of the shared β 1 chain **results in** simultaneous inability to assemble functional **IL-12 receptors** (IL-12R β 1/ β 2) and **IL-23 receptors** (IL-12R β 1/IL-23R) on T and NK cells.
3. **Branch A (IL-12 arm):** Absent IL-12 signaling **leads to** failure of STAT4 activation in T/NK cells → markedly reduced **IFN- γ production** (PMID: 32025907).

4. Deficient IFN- γ **results in** impaired activation of infected **macrophages** and defective killing of intracellular mycobacteria and *Salmonella* → poorly controlled, potentially disseminating infection with weakly virulent mycobacteria and non-typhoidal *Salmonella* (PMID: 25453225).
5. **Branch B (IL-23 arm):** Absent IL-23 signaling **leads to** impaired **IL-17/Th17 immunity** at mucosal surfaces → **mucocutaneous candidiasis** in ~25% of patients (inferred mechanism, strongly supported) (PMID: 24186907).
6. **Residual protection (why penetrance is incomplete):** Partly IFN- γ -independent and IL-12/IL-23-independent antimycobacterial pathways **provide** residual immunity, so a substantial fraction of genotype-positive individuals remain asymptomatic and mycobacterial disease seldom recurs (demonstrated in patients; dissected in *Il12rb1*-KO mice) (PMID: 35891311, PMID: 21057261).



Molecular pathway (upstream → downstream). IL-12/IL-23 → IL-12Rβ1 (receptor) → JAK–STAT4 → **IFN- γ** → IFN- γ R/STAT1 → macrophage antimycobacterial program. The lesion is **upstream**, at the receptor level of IFN- γ *production*; the downstream IFN- γ *response* machinery is intact — which is precisely why exogenous recombinant IFN- γ can bypass the block (PMID: 25453225).

Classification. *IL12RB1* belongs to the MSMD genes that impair the **production** of IFN- γ : "These disorders impair the production of (*IL12B*, *IL12RB1*, *IRF8*, *ISG15*, *NEMO*) or the response to (*IFNGR1*, *IFNGR2*, *STAT1*, *IRF8*, *CYBB*) IFN- γ " (PMID: 25453225). Recent genetics show "both *IL-12* and *IL-23* are required for optimal levels of IFN- γ " (PMID: 32025907).

Immune system involvement. This is a pure **immunodeficiency** (not autoimmune/autoinflammatory) of cell-mediated, granulomatous antimycobacterial immunity. Cell types involved: **T lymphocytes (CL:0000084)**, **natural killer cells (CL:0000623)**, **Th17 cells (CL:0000899)**, and **macrophages (CL:0000235)**. Biological processes: IFN- γ production (GO:0032609), positive regulation of IFN- γ production (GO:0032729), IL-12-mediated signaling (GO:0035722), IL-23-mediated signaling (GO:0038155), T-helper 17 cell differentiation (GO:0072539), macrophage activation involved in immune response (GO:0002281). Cellular component: plasma-membrane cytokine receptor complex (GO:0005886).

Biomarker note. Serum **neopterin remains elevated** even with complete absence of IFN- γ activity, so neopterin cannot be used to diagnose IFN- γ /IL-12/IL-23-pathway MSMD (PMID: 16339068).

7. Anatomical Structures Affected

- **Primary body system:** immune/hematopoietic system (UBERON:0002405 immune system).
- **Lymphoreticular:** lymph nodes (UBERON:0000029) — lymphadenitis, especially axillary post-BCG; spleen (UBERON:0002106); liver (UBERON:0002107) — hepatomegaly/hepatosplenic dissemination.
- **Skin and subcutaneous tissue** (UBERON:0002097) — BCG injection-site nodules, fistulae, cutaneous NTM disease.
- **Lung** (UBERON:0002048) — mycobacterial pulmonary disease.
- **Bone marrow / disseminated reticuloendothelial involvement** in severe cases.
- **Oropharyngeal mucosa** (UBERON:0000167) — candidiasis.
- **Cell/tissue level:** macrophages within granulomas; T and NK lymphocytes (functional defect, not structural). **Subcellular:** plasma membrane (site of the missing receptor); JAK–STAT signaling in the cytosol.
- **Lateralization:** typically follows inoculation/dissemination pattern — e.g., ipsilateral (to BCG injection) axillary lymphadenitis initially, later bilateral/disseminated.

8. Temporal Development

- **Onset:** pediatric, typically **infancy to early childhood**; mean age at first infection **2.4 years**, frequently precipitated by BCG vaccination in the first months of life ([PMID: 21057261](#)). Onset pattern is usually **subacute** (progressive lymphadenitis/nodule) but can be acute-disseminated.
- **Progression:** variable — local disease may resolve with therapy, or progress to disseminated, life-threatening infection. Mycobacterial disease characteristically **does not recur** after cure; *Salmonella* recurs more often ([PMID: 12591909](#), [PMID: 23429356](#)).
- **Course:** episodic infections superimposed on a **chronic lifelong** underlying susceptibility. Susceptibility tends to attenuate with age (adults are less prone to new mycobacterial episodes).
- **Critical period:** the peri-vaccination window in infancy is the key vulnerable/interventional period — avoiding live BCG prevents the commonest trigger.

9. Inheritance and Population

- **Inheritance:** **autosomal recessive** (complete deficiency requires biallelic LOF).
- **Penetrance:** **incomplete / low** — 8/29 genetically affected siblings (27%) remained asymptomatic in the international survey ([PMID: 21057261](#)); most MSMD defects "*do not show complete clinical penetrance*" ([PMID: 25453225](#)).
- **Expressivity:** highly **variable**, from asymptomatic to fatal disseminated disease ([PMID: 24064560](#)).
- **Founder effects / consanguinity:** major drivers; recurrent country-specific alleles and high parental consanguinity in cohorts from Iran, Turkey, and the Middle East ([PMID: 32602053](#), [PMID: 30968642](#)); founder effects are explicitly documented for the near-phenocopy IL-12p40 deficiency ([PMID: 23429356](#)).
- **Epidemiology:** overall a **rare** disorder; precise prevalence/incidence are not established, but IL-12R β 1 deficiency is the **most common single genetic etiology of MSMD** ([PMID: 21057261](#), [PMID: 36044171](#)). Diagnoses cluster in **BCG-vaccinating, high-consanguinity** populations.

- **Sex ratio / age distribution:** no strong sex bias intrinsic to the recessive disorder; cohort male predominance (e.g., 64% male in the Mexican cohort) likely reflects ascertainment (PMID: 36044171). Age distribution skews pediatric.
- **Anticipation / mosaicism:** not features of this disorder.

10. Diagnostics

Key caveat: patients are "*otherwise healthy individuals with no overt abnormalities in routine hematological and immunological tests*" (PMID: 25453225). Routine CBC, immunoglobulins, and lymphocyte subsets are typically normal — so a **high index of suspicion** and **specific testing** are essential.

Functional immunology (screening). - Whole-blood/PBMC stimulation with **BCG ± IL-12 and BCG ± IFN-γ**, measuring IFN-γ and IL-12p40 by ELISA: deficient IFN-γ production in response to IL-12 points to the IL-12/IL-12R arm (PMID: 31158284, PMID: 17120032). - **Flow cytometry** for IL-12Rβ1 surface expression on activated T cells / EBV-transformed B-cell lines: **absent expression** confirms complete deficiency (PMID: 19099833, PMID: 31158284). - **Do NOT rely on serum neopterin** — it stays elevated despite absent IFN-γ activity (PMID: 16339068).

Genetic testing (confirmatory). Sanger sequencing of *IL12RB1*, targeted **immunodeficiency NGS panels**, or **whole-exome/whole-genome sequencing** identify biallelic pathogenic variants (PMID: 31158284, PMID: 32602053). **Dried blood spot** testing has enabled diagnosis in resource-limited/refugee settings (PMID: 30740107). Carrier testing and prenatal diagnosis are feasible once the familial variant is known (PMID: 19099833).

Microbiology / pathology. Culture and molecular identification of mycobacteria (BCG strain confirmation) or *Salmonella*; biopsy typically shows **granulomatous inflammation** (caseous or poorly formed granulomas).

Differential diagnosis. Other MSMD genes (*IL-12p40/IL12B* — a near-phenocopy; *IFNGR1/2*, *STAT1*, *ISG15*, *IRF8*, *NEMO*, *TYK2*), chronic granulomatous disease (CGD), SCID, combined immunodeficiencies, and other primary immunodeficiencies presenting with BCG disease (PMID: 34780073, PMID: 10959079). Distinguishing IL-12Rβ1 from IFN-γR defects matters prognostically: IL-12Rβ1 has better outcomes and non-recurring mycobacterial disease.

Screening. In families with a known proband, **cascade genetic testing** of siblings and **prenatal/newborn** targeted testing guide BCG-avoidance and early surveillance (PMID: 19099833).

11. Outcome / Prognosis

Comparatively favorable relative to other MSMD etiologies.

Cohort	N (IL12RB1 / MSMD)	Key outcome
International survey (PMID: 21057261)	141 patients / 102 kindreds	70% survival ; mean follow-up 12.7 ± 9.8 yr; 27% of affected sibs asymptomatic
Earlier cohort (PMID: 12591909)	41 patients	Only 5 childhood deaths; mycobacterial infections did not recur
Mexico (PMID: 36044171)	22 MSMD (13 IL12RB1)	7 deaths from disseminated BCG
Shanghai 15-yr PID/BCG cohort (PMID: 41748971)	109 PID w/ BCG disease (MSMD 43.1%)	5-yr survival 80.3%, 10-yr 69.3%; HSCT success 75.8% vs 0%

Mortality is driven principally by **disseminated BCG disease**, especially where diagnosis is delayed (PMID: 36044171). Prognostic factors include severity/dissemination at presentation, timeliness of diagnosis and antimycobacterial therapy, access to recombinant IFN- γ , and HSCT for refractory disease. Mycobacterial **recurrence is uncommon**, a favorable prognostic feature; recurrent salmonellosis is a recognized complication.

12. Treatment

Pharmacotherapy — **pathogen-directed**. - **Prolonged multidrug antimycobacterial therapy** for BCG/NTM/TB (regimens per species and susceptibility; drug-resistant disease may require individualized regimens) (PMID: 33631907). - **Antibiotics** for *Salmonella* (with attention to recurrence). - **Antifungals** for candidiasis.

Immunomodulation — **recombinant human IFN- γ (rhIFN- γ , NCIT: Recombinant Interferon Gamma)**. Because the defect impairs IFN- γ *production* while the IFN- γ *response* pathway is intact, **exogenous IFN- γ bypasses the block**. In the Chinese cohort, "77.8% of

patients received *rhIFN- γ* treatment, which can improve the prognosis of patients with *IL12RB1* deficiency" (PMID: 31367980). Combined antimycobacterial + IFN- γ therapy achieved control in individual refractory cases, with IFN- γ maintenance (PMID: 41668770, PMID: 30968642).

Hematopoietic stem cell transplantation (HSCT). Curative for the underlying immune defect and highly effective against mycobacterial disease in severe/refractory cases: "*Patients who received hematopoietic stem cell transplantation therapy (HSCT) had a significantly higher success rate with antimycobacterial treatment (75.8% vs. 0%)*" (PMID: 41748971). HSCT is reserved for severe/relapsing disease given the generally favorable natural history of many IL-12R β 1 patients.

Supportive / surgical. Drainage or excision of suppurative lymphadenitis; nutritional support; management of disseminated organ involvement.

Pharmacogenomics / gene & RNA therapies. No approved gene, cell (beyond HSCT), or RNA therapy specific to *IL12RB1*; gene correction is conceptually attractive (proof-of-concept complementation restores function in vitro, PMID: 11424023) but not clinically available.

Suggested NCIT intervention terms: Recombinant Interferon Gamma; Hematopoietic Stem Cell Transplantation; Antimycobacterial/Antitubercular Agents; Antibiotic Therapy; Antifungal Agent.

13. Prevention

- **Primary prevention — avoid live BCG vaccine** in individuals/families known or suspected to have MSMD; BCG is the single most common disease trigger (PMID: 21057261, PMID: 23429356). This is the central actionable prevention measure, especially in high-TB-burden countries with universal neonatal BCG (PMID: 41668770).
- **Secondary prevention — early recognition and treatment.** Suspect an inborn error of immunity in any child with disseminated BCG/NTM or non-typhoidal *Salmonella* disease and refer for early immunological evaluation (PMID: 33631907, PMID: 41771439).
- **Genetic counseling / screening.** Autosomal-recessive risk counseling; **cascade testing** of siblings; **carrier and prenatal testing** once the familial variant is known (PMID: 19099833); consanguinity counseling. Dried-blood-spot testing extends screening to resource-limited settings (PMID: 30740107).
- **Tertiary prevention.** Antimicrobial prophylaxis/surveillance for recurrent salmonellosis; prompt treatment of new infections; consideration of HSCT in severe/refractory disease.

- **Prophylaxis / immunization note.** No vaccine prevents the disorder; live vaccines are contraindicated. Adjunctive rhIFN- γ can be used therapeutically and, in effect, prophylactically as maintenance in refractory disease ([PMID: 41668770](#)).

14. Other Species / Natural Disease

- **Human:** *Homo sapiens* (NCBI:txid9606). No naturally occurring companion-animal or wildlife counterpart of *IL12RB1* deficiency is documented in the reviewed literature (no OMIA entry identified for this specific defect).
- **Orthologue:** mouse *Il12rb1* (NCBI Gene ID 16161), highly conserved; the IL-12/IL-23 $\rightarrow$ IFN- γ axis is evolutionarily conserved across mammals, underpinning the validity of the mouse model.
- **Zoonotic / cross-species transmission:** not applicable — this is a host genetic susceptibility, not a transmissible disease. (The triggering pathogens, e.g., *M. bovis*, do have zoonotic relevance, but the disorder itself is non-transmissible.)

15. Model Organisms

- **Primary model: *Il12rb1*-deficient (knockout) mouse.** It recapitulates the core human phenotype: "*Interleukin-12 receptor β 1 (IL12RB1)-deficient individuals show increased susceptibilities to local or disseminated BCG infection and environmental mycobacteria infection,*" and the model is used specifically to probe why "*the low clinical penetrance of IL12RB1 deficiency and low recurrence rate of mycobacteria infection suggest that protective immunity still exists in this population*" ([PMID: 35891311](#)).
- **Model type:** mammalian germline knockout (constitutive). Useful for dissecting IFN- γ -dependent vs IFN- γ -independent antimycobacterial immunity, disseminated BCG pathogenesis, and correlates of residual protection.
- **Phenotype recapitulation:** good face validity for **disseminated BCG susceptibility**; captures the impaired IL-12 $\rightarrow$ IFN- γ axis and the residual-immunity phenomenon.
- **Limitations:** mice are not routinely BCG-vaccinated as neonates (human trigger context differs); the candidiasis/mucosal IL-17 phenotype and the full human infection spectrum are not fully modeled; genetic background modifies penetrance.

- **In vitro / cellular models:** patient T cells and EBV-transformed B-cell lines used for expression and complementation studies — wild-type *IL12RB1* transfection restores expression and IL-12 responsiveness ([PMID: 11424023](#)); flow-cytometry and cytokine-stimulation assays on patient PBMCs ([PMID: 31158284](#)).
- **Resources:** MGI (mouse *Il12rb1*); patient-derived cell lines from reference immunology laboratories.

Mechanistic Model / Interpretation

The disorder is best understood as a **single upstream lesion with two downstream branches plus a safety net**:

Element	Consequence	Clinical readout
Loss of IL-12R β 1 (shared chain)	No IL-12 signaling $\rightarrow$ $\downarrow$ STAT4 $\rightarrow$ $\downarrow$ IFN-γ	Macrophages fail to kill mycobacteria/ <i>Salmonella</i> $\rightarrow$ disseminated infection
Loss of IL-12R β 1 (shared chain)	No IL-23 signaling $\rightarrow$ $\downarrow$ Th17/IL-17	Mucocutaneous candidiasis (~25%)
Intact IFN- γ response machinery	Exogenous IFN- γ still works	rhIFN- γ therapy is effective
Redundant IFN- γ -independent immunity	Residual mycobacterial control	Low penetrance; mycobacterial disease rarely recurs; ~70% survival

This model coherently explains (a) the **narrow pathogen spectrum** (IL-12 is redundant for most microbes; [PMID: 12591909](#)), (b) the **normal routine immune tests** (the defect is a specific cytokine-receptor lesion, not a global immune failure), (c) the **therapeutic logic of rhIFN- γ** (production defect, response intact), and (d) the **favorable prognosis and incomplete penetrance** (residual immunity). It also predicts the **close phenocopy with IL-12p40/IL12B deficiency**, since both remove IL-12(IL-23)-driven IFN- γ ([PMID: 10959079](#), [PMID: 23429356](#)).

Evidence Base

PMID	Contribution	Supports
21057261	International 141-patient survey — most common MSMD etiology; 70% survival; 27% asymptomatic sibs; organism spectrum	F001 (epidemiology, prognosis, penetrance)
12591909	Low penetrance, broad resistance, favorable outcome; IL-12 redundant except mycobacteria/ <i>Salmonella</i>	F001 (selective susceptibility)
11424023	Complementation proof — WT <i>IL12RB1</i> restores expression/function	F002 (causality, LOF)
24186907	~25% candidiasis via impaired IL-23/IL-17	F002 (IL-23 branch)
32025907	Both IL-12 and IL-23 required for optimal IFN- γ	F002 (pathway)
31367980	rhIFN- γ improves prognosis (Chinese cohort)	F003 (treatment)
41748971	HSCT antimycobacterial success 75.8% vs 0%; survival curves	F003 (treatment)
10959079	Places IL-12R β 1 in the IFN- γ /IL-12 circuit; pathogen spectrum	F004 (classification)
23429356	IL-12p40 phenocopy; founder effects; recurrence pattern	F004 (founder effects, differential)
35891311	<i>Il12rb1</i> -KO mouse recapitulates disseminated BCG; residual immunity	F005 (model, penetrance)
29256176	Human variants abolish surface expression and IL-12 response	F005 (functional genetics)
25453225	Production-vs-response classification; normal routine tests; broad spectrum	F006 (diagnosis, mechanism)
16339068	Neopterin remains elevated — not a usable diagnostic marker	Diagnostics caveat
31158284	Functional flow + stimulation assays; novel variants	Diagnostics
36044171		

PMID	Contribution	Supports
	Mexican cohort — IL12RB1 leading MSMD cause; BCG-driven mortality	Epidemiology/prognosis
32602053	Iranian cohort — consanguinity, allelic heterogeneity	Population genetics
30968642, 30740107, 33631907, 41668770, 41771439, 19099833, 24064560, 23726680, 34780073	Case reports/cohorts — variant types, DBS diagnosis, IFN- γ maintenance, Salmonella presentation, prenatal diagnosis, variable penetrance, NTM disease, BCG differential	Phenotypes, variants, treatment, prevention

Evidence source types: human clinical cohorts and case reports (majority); in vitro functional/complementation studies (PMID: 11424023, PMID: 31158284); model-organism (mouse) study (PMID: 35891311).

Limitations and Knowledge Gaps

1. **No precise prevalence/incidence figures.** IL-12R β 1 deficiency is established as the most common MSMD etiology but exact population rates are unquantified; ascertainment is biased toward BCG-vaccinating, high-consanguinity regions.
2. **Penetrance mechanism incompletely defined.** The identity of the residual IFN- γ -independent protective pathway(s) is unresolved; the mouse model is actively being used to address this (PMID: 35891311).
3. **No standardized quality-of-life or disability outcome data** (EQ-5D/SF-36/PROMIS) for this disorder.
4. **Genotype–phenotype correlation is weak/absent.** Complete-null alleles produce variable severity; no consistent modifier gene is identified.
5. **Treatment evidence is largely observational** (cohorts/case series), not randomized; optimal rhIFN- γ dosing/duration and precise HSCT indications remain to be standardized.
6. **Candidiasis–IL-17 link is well-supported but mechanistically inferred** rather than directly proven in each patient.

Proposed Follow-up Experiments / Actions

1. **Define residual immunity:** use the *Il12rb1*-KO mouse and patient cells to map the IFN- γ -independent antimycobacterial pathway(s) that explain low penetrance and non-recurrence (single-cell transcriptomics of granuloma macrophages; IL-12/IL-23-independent IFN- γ sources).
 2. **Registry and natural-history study:** establish a prospective international registry to quantify prevalence, long-term outcomes, and standardized QoL measures.
 3. **Genotype-modifier discovery:** whole-genome sequencing of concordant/discordant sib pairs to identify penetrance modifiers.
 4. **Therapeutic optimization:** prospective evaluation of rhIFN- γ dosing/duration and clearer HSCT-selection criteria; explore gene-correction (autologous HSC + *IL12RB1* repair) building on in vitro complementation proof-of-concept.
 5. **Screening policy:** evaluate targeted pre-BCG screening (cascade genetic testing, dried-blood-spot assays) in high-consanguinity, high-TB-burden populations to prevent BCG disease.
 6. **Biomarker development:** validate IL-12-response functional assays and flow-cytometric IL-12R β 1 expression as standardized frontline diagnostics (and formally deprecate neopterin for this purpose).
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Report compiled from 6 confirmed findings and 30 reviewed papers across the autonomous discovery iterations. All mechanistic and clinical claims are cited to primary literature by PMID.
